

CSA0711
SIMATS ENGINEERING
ASSESSMENT TOOL 3

26

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I P. Sashan (192423942) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sashan

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	1	1	1
Debugging	25%	2	2	2	2	2
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1/2	1/2	1/2	1/2	1/2

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Debugging an IPv4 Workstation Network Configuration

Problem Identification

A network administrator has configured a workstation with the following network settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26. This indicates that there is a network configuration error preventing successful communication. The objective is to identify the root cause of the problem, verify whether the IP address and gateway belong to the correct subnet.

Systematic Debugging Process

Analyze the given subnet

255.255.255.192

This is equivalent to /26.

A /26 subnet contains:

Usable host addresses: $2^{(32-26)} = 64$ addresses

$$64 - 2 = 62$$

Each subnet increases in blocks of 64.

Identify the subnet of the Configured IP Address

Configured IP Address:

192.168.10.65

This address falls within:

192.168.10.64 - 192.168.10.127

Network Address:

192.168.10.64

Valid Host Range

192.168.10.65 - 192.168.10.126

The configured workstation belongs to the subnet:

192.168.10.64/26

Verify the Default Gateway

192.168.10.1

The gateway 192.168.10.1 belongs to the subnet:

192.168.10.0/26

Network Address:

192.168.10.0

Broadcast Address:

192.168.10.63

Valid Host Range:

192.168.10.1 - 192.168.10.62

Identify the Configuration Errors

The debugging process reveals the following errors:

The workstation IP address

192.168.10.65

belongs to the subnet

192.168.10.64/26.

The default gateway

192.168.10.1

belongs to the subnet

192.168.10.0/26

→ Communication with other hosts fails.

→ Internet Connection Fails

→ Access to resource outside the local subnet becomes impaired
Keep the existing gateway.

192.168.10.1

Assign the workstation an IP address with

192.168.10.0/26.

② Debugging an IPv6 Network Configuration

Problem Identification

Two devices in an IPv6 network are configured with addresses:

- Device A: 2001:dbf:PF00:42:9329/64
- Device B: 2001:dbf:001::1/64

Although both devices are physically connected to the same network, they are unable to communicate with each other. The purpose of this debugging task is to identify the root cause of the communication failure by examining the IPv6 addressing configuration, verifying the network prefix, identifying any addressing mismatches, and optimizing the configuration to ensure successful communication.

Systematic Debugging Process

Expand the IPv6 Address

IPv6 addresses are often written in compressed form by removing leading zeros and replacing consecutive zero blocks with a double colon (::). During debugging, it is important to expand them into their full 32-bit hexadecimal notation.

Device A

2001:db8:ff00:42:9329

Expanded Address:

2001:0db8:0000:0000:0000:ff00:0042:9329

Device B

Compressed Address:

2001:db8:0:0:1::1

Expanded Address:

2001:0db8:0000:0000:0001:0000:0000:0001

Determine the Network Prefix

A /64 prefix means:

- First 64 bits = Network Prefix
- Last 64 bits = Interface Identifier

Device A

Address:

2001:0db8:0000:0000:0000:ff00:0042:9329

Network Prefix:

2001:0db8:0000:0000::/64

Device B

Address:

2001:db8:0000:0000:0001:0000:0000:0001

Network Prefix:

2001:db8:0000:0000::/64

Step 3: Verify whether both devices belong to the same Subnet

Comparing the network prefixes:

Device

network prefix

Device A

2001:db8:000::/64

Device B

2001:db8:0:0::/64

Both devices belong to the same /64 subnet because the first 64 bits are identical.

Step 4: Identify the Addressing or Prefix Mismatches

- Incorrect IPv6 default gateway configuration
- IPv6 disabled on one of the devices
- Firewall blocking ICMPv6 traffic
- Duplicate IPv6 addresses
- Incorrect neighbor discovery protocol (NDP) configuration
- Missing route advertisement

Analyzing and optimizing IPv4 subnet Allocation using VLSM

Problem Identification

An organization has been allocated the IPv4 network 192.168.1.0/24 for its Internal Communications. The network has been divided into multiple departmental subnets using the subnet masks 126, 127 and 128. However, some departments report that they do not have enough IP addresses for their devices, while other departments have a large number of unused IP addresses. This indicates that the current subnet allocation is inefficient and does not match the actual requirements of each department.

Systematic Debugging Process

Step 1: Analyze the Existing Subnets

The organization uses the following subnet masks:

- Department A → 126
- Department B → 128
- Department C → 129

Department D (123)

Subnet Mask:

255.255.255.224

Total Addresses:

$$2^{(32-27)} = 32$$

Usable Hosts:

$$32 - 2 = 30$$

Network Address: 192.168.1.64

Broadcast Address: 192.168.1.95

Valid Host Range: 192.168.1.65 - 192.168.1.94

Department C (125)

Subnet Mask:

255.255.255.240

Total Addresses:

$$2^{(32-28)} = 16$$

Usable Hosts:

$$16 - 2 = 14$$

Network Address: 192.168.1.96

Broadcast Address: 192.168.1.111

Valid Host Range: 192.168.1.97 - 192.168.1.110

Conclusion

The subnet allocation analysis identified that the existing addressing scheme did not efficiently utilize the available IPv4 address space.

④ Debugging an IPv6 Network Configuration

Problem Identification

Two devices in an IPv6 network have been configured with following addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to same physical network they are unable to communicate with each other. IPv6 network successful communication requires that devices share the same network prefix and have unique interface identifiers.

This task involves debugging the IPv6 configuration by examining the compressed addresses, determining the network prefix, checking whether both devices belong to the same subnet, identifying the cause of the communication failure, optimizing the addressing configuration, and calculating the number of

Calculate the Number of Available Host Address.

IPv6 addresses contain:

128 bits

Prefix length:

/64

Host Portion:

$$128 - 64 = 64$$

Available Interface addresses:

$$2^{64} = 18,446,744,073,709,551,616$$

Therefore

Available Host Address

$$18,446,744,073,709,551,616$$

⑤ Debugging a Route Routing Table

Problem Identification

A router is configured with the following routing table:

Destination network:

192.168.1.0/24

192.168.0/24

Default Route (0.0.0.0)

Next Hop

10.0.0.2

10.0.0.3

10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended destination network. The router should examine its routing table, identify the correct destination network, and forward the packet to the appropriate next hop router. However, the communication failure suggests that there may be a routing configuration problem.

The objective of this task is to debug the routing table by identifying the mismatched routing entry, determining the selected next hop router.

Determine the Default Route

If a packet arrives with a destination address that does not match any routing table entry, the router forwards the packet using the default route.

Default Route:

0.0.0.0/0

Next Hop Route:

10.0.0.1

Therefore,

all unknown destination networks are forwarded to

Default Gateway

10.0.0.1